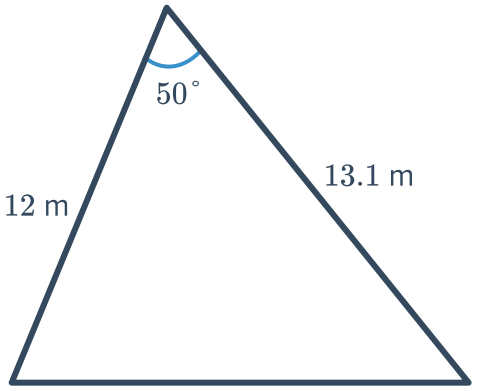
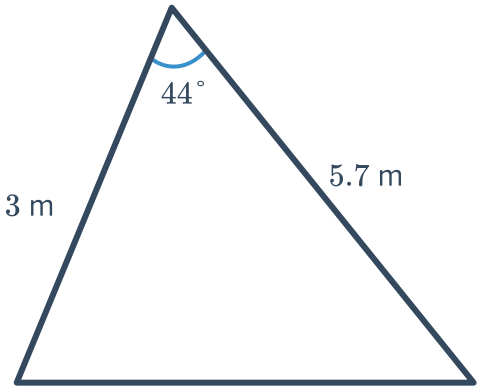
Area of non-right triangles

1 Calculate the area of the following triangles, rounded to two decimal places:

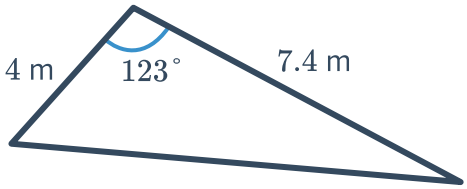
a



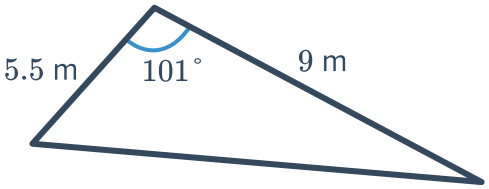
b



c



d



2 Find the area of the following triangles, to one decimal place:

a Sides lengths of $9.5\ \text{ft}$, $10\ \text{ft}$ with an included angle measuring 47°.

b Sides lengths of $18.4\ \text{in}$, $20.5\ \text{in}$ with an included angle measuring 99°.

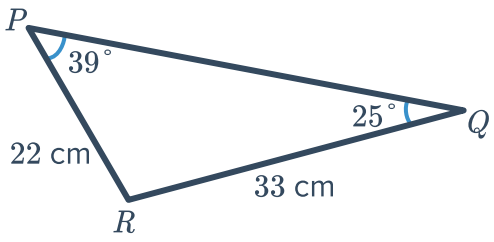
3 Find the exact area of an equilateral triangle with a side length of $6\ \text{yd}$.

4 Find the exact area of an equilateral triangle with a perimeter of $24\ \text{in}$.

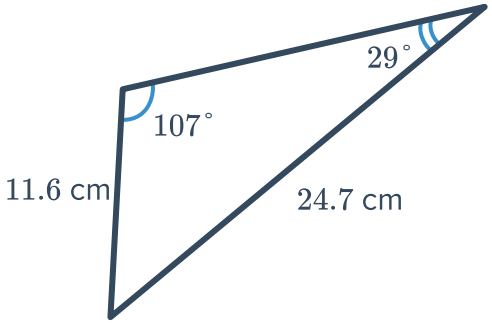
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5 Calculate the area of the following triangles, to the nearest square centimeter:

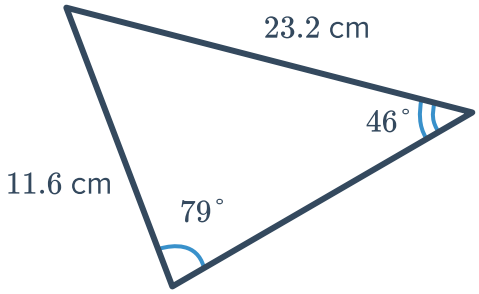
a



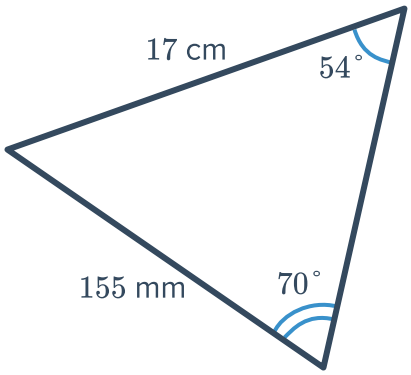
b



c



d

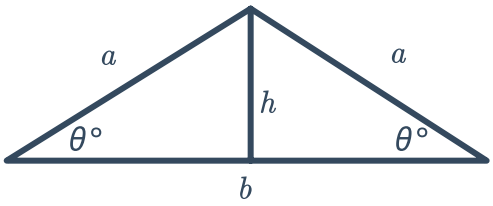


6 Consider the diagram of an isosceles triangle where h is the height perpendicular to base, b:

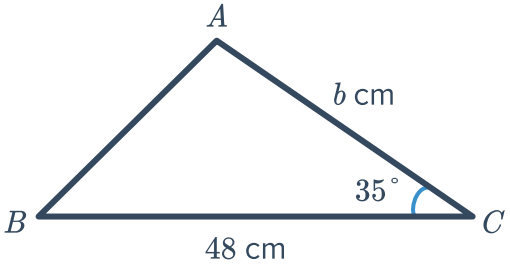
a Form an expression for h in terms of θ and a.

b Find b in terms of θ and a.

c Form an expression for the area A of the larger triangle, in terms of θ and a.

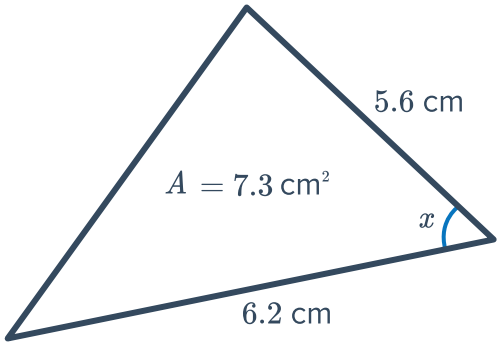


7 The following triangle has an area of $520\ \text{cm}^2$. Find the length of side b. Round your answer to the nearest centimeter.



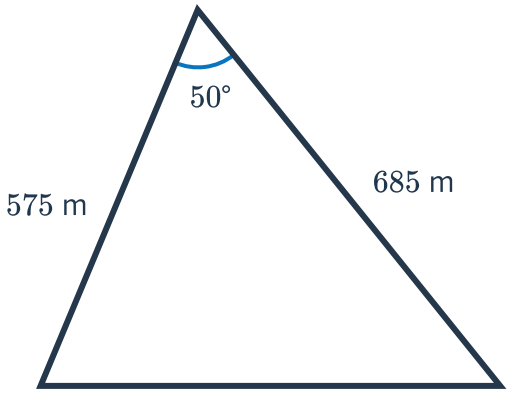
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8 Consider the diagram and find the measure of angle x, correct to one decimal place:



Applications

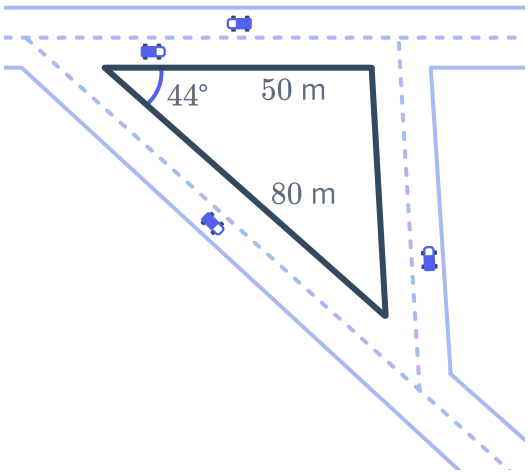
9 A triangular paddock has measurements as shown in the diagram:



a Find the area of the paddock. Round your answer to the nearest square meter.

b State the area in hectares. Round your answer to two decimal places.

10 An industrial site in the shape of a triangle is to take up the space between where three roads intersect as shown in the diagram.
Find the area of the site. Round your answer to two decimal places.



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11 The Bermuda Triangle is an area in the Atlantic Ocean where many planes and ships have mysteriously disappeared. Its vertices are at Bermuda (B), Miami (M) and Puerto Rico (P):



Find the area of the Bermuda Triangle. Round your answer correct to two decimal places.

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